SOHEE KIM

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SUMMARY

Ph.D. student in KAIST-KIST AI Robotics and Student Researcher at KIST, with nearly five years of industry experience at Hyundai Motor Company's Robotics Lab. Experienced in **keypoint detection, human motion analysis, and action recognition**, with **ROS and TensorRT** deployment skills. Research interests: **VLA and WAM** for autonomous robotics.

EDUCATION

AUG.2026- Present	**KAIST** KAIST-KIST AI Robotics Ph.D. Student Advisor: Professor SeungRyong Kim	**Seoul, Korea**
MAR.2020- FEB.2022	**KyungHee University** Department of Computer Science and Engineering M.S. in Computer Science and Engineering Advisor: Professor SeungKyu Lee GPA: 4.3/4.3	**Seoul, Korea**
MAR.2016- FEB.2020	**KyungHee University** Department of Computer Science and Engineering B.S. in Computer Science and Engineering GPA: 3.93/4.3 **Honors: Valedictorian in SW College (Rank: 1)**	**Seoul, Korea**

INTERNATIONAL CONFERENCES/JOURNALS

1. **SH Kim**, SK Lee. "Continual Learning with Neuron Activation Importance", In Springer International Conference on Image Analysis and Processing (ICIAP) 2022, Lecce, Italy **(Oral Presentation)**
   This work was based on my Master's thesis.
2. **SH Kim**, SK Lee. "Occlusion Robust Part-aware Object Classification through Part Attention and Redundant Features Suppression", In ACM SIGGRAPH ASIA 2021 Posters (Poster Paper), Tokyo, Japan

DOMESTIC CONFERENCES/JOURNALS

1. **SH Kim**, SK Lee. "Self-Occlusion Robust Sign Language Recognition using Skip LSTM", In Korea Computer Congress (KCC) Undergraduate/Junior Paper 2019, Jeju, Korea
2. **SH Kim**, SK Kim, "Sparse Depth Noise-Guided Diffusion Model for Indoor Depth Completion", Hyundai Motor Company Conference 2024, Korea

PATENT APPLICATIONS

1. "Supporting Method of Motion Recognition for Robot, Computing Device and System Supporting the same", Patent Applications filed in Korea, the United States, and China
2. "Action Recognition Apparatus and Method," Patent Applications filed in Korea, the United States, China, and Germany
3. "Method and Device for Reading Gauges", Patent Applications filed in Korea and the United States
4. "Method and Device for Generating Depth Map", Patent Applications filed in Korea, the United States, China, and Germany
5. "Method and Device for Generating Depth Map", Patent Applications filed in Korea, the United States, China, and Japan
6. "Method and Apparatus for Training Pose Estimation Model using Heterogeneous Training Dataset", Patent Applications filed in Korea
7. "Method and Apparatus for Evaluating Human Body Load based on Image", Patent Applications filed in Korea

PROFESSIONAL EXPERIENCE

Sep.2026-Present

VLA Research,
Center for AI Research, Korea Institute of Science and Technology (KIST)
Advisor: HwaSup Lim

- Conducting research on **Vision-Language-Action (VLA)** models.

Dec.2025-Aug.2026

Physics-based Humanoid Control via Video-to-Motion Imitation Learning,
Robotics Recognition AI Team, Hyundai Motor Company

- Developed a video-to-motion pipeline generating SMPL-based motion data from monocular video, addressing temporal consistency issues in reconstructed motion sequences.
- Trained physics-based humanoid control policies using diffusion-based imitation learning on reconstructed motion trajectories.

July.2025-Nov.2025

Multi-Class Object Keypoint Detection Algorithm Development,
Robotics Vision AI Team, Hyundai Motor Company

- Extended human keypoint detection algorithms to a **multi-class object keypoint detection** framework for diverse robotic applications.
- Integrated into the **MobED** robot platform and demonstrated at **IREX 2025** (International Robot Exhibition) and **CES 2026** (Consumer Electronics Show).

Apr.2024-Aug.2026

Human Keypoint Detection Algorithm Development,
Robotics Vision AI Team, Hyundai Motor Company

- Developed 2D and 3D human keypoint detection algorithms for analyzing joint angles and assessing worker joint load.
- **Deployed in the mass-produced X-ble Shoulder** wearable robot.
- Built a custom dataset using **Qualisys** motion capture system and **QTM** software, integrating both custom and public datasets with varying skeletal configurations.
- Applied advanced augmentation techniques, including camera zooming and keypoint occlusion augmentations, to enhance model robustness.
- Built a web interface to demonstrate live algorithm performance and generate ergonomic assessment reports.

Aug.2023-Aug.2026

Action Recognition Algorithm Development,
Robotics Vision AI Team, Hyundai Motor Company

- Developed action recognition from 2D keypoints and velocity features; deployed in C++ with **TensorRT** for real-time inference.
- Piloted action-mimicking robot services and CCTV-based user intent analysis.

Jun. 2023

CVPR 2023 Robotics Vision AI Team Representative,
R&D Division, Hyundai Motor Company

- Presented core technologies and research outcomes of the team to international researchers.
- Conducted recruitment promotion and technical discussions in English with global participants.

Apr. 2022-May. 2023

Anomaly Detection Algorithm Development for Patrol Robots,
Robotics Vision AI Team, Hyundai Motor Company

- Developed vision-based algorithms for autonomous patrol robots to detect abnormal equipment conditions in smart factories.
- Detected gauge anomalies through keypoint detection on analog meter needles.
- Identified valve malfunction by classifying rotational states.
- Recognized work status by classifying lamp colors.
- Integrated algorithms into the robot control system using **ROS1** for real-time operation.
- Conducted a PoC using a mobile robot for over six months at the Hyundai Motor Group Innovation Center in Singapore (**HMGICS**), validating the algorithms' industrial feasibility.

AWARDS & HONORS

2020-2022	Entrance Examination Scholarship **(full tuition)**, KyungHee University, Korea
2020	Graduated as Valedictorian **(Rank :1)**, College of Software, KyungHee University, Korea
2019	Best Excellence Award in KHU SW-Festival, KyungHee University, Korea
2019	Popularity Award in KHU SW-Festival, KyungHee University, Korea
2019	Special Prize in KHUTHON, KyungHee University, Korea
2019-1	SW Center College Scholarship, KyungHee University, Korea
2018-2	Bright Society Scholarship (International Training Program)
2018-1, 2017-1	Humanitas College Scholarship, KyungHee University, Korea
2017-2	Superiority Scholarship (Quarter tuition), Khunghee University, Korea
2016-2	Superiority Scholarship (**full tuition**), Kyunghee University, Korea

ACADEMIC PROGRAMS

JAN.2020 **Global Academic and Industry Experience Program | CES & UCSD, USA**

University-selected Participant

- Selected as one of the top undergraduate students to participate in a global academic–industry immersion program.
- Attended **CES 2020** to analyze state-of-the-art AI trends and industry applications.
- Participated in a winter program at **UC San Diego**, including lab tours, seminars, and a one-day collaborative project.

EXTRA CURRICULAR ACTIVITIES

2016-present **Street Dance Club (BMB), KyungHee University**

- Served as Urban Dance Team Leader (2017).
- Choreographed original routines and led bi-weekly group practice sessions.
- Participated in regular performances and college festivals until 2023.

2018 **Teacher, After-School Program – SeoCheon Elementary School**

- Taught Scratch programming to elementary school students.

2018-1, 2017-1 **Mentor, Freshmen Seminar - KyungHee University**

- Led a semester-long social project as part of the seminar program.
- Provided weekly counseling and guidance to first-year students for academic and social adjustment.
- **Awarded 1st prize** in the seminar project for supporting single mothers (2018).